LESSON 16: ROUTER OPERATION, STATIC ROUTES, AND DEFAULT ROUTES

TRAINING FOCUS

This lesson is part of the FEMTECT Network Engineering Foundation course. It is written to help students build strong networking fundamentals through clear explanations, real enterprise context, practical troubleshooting logic, and certification-aware technical depth.

VISUAL DIAGRAM: ROUTE SELECTION LOGIC

Visual flow

From	Relationship	To
Packet arrives at router	connects to	Check routing table
Check routing table	connects to	Matching routes found?
Matching routes found?	No	Drop packet
Matching routes found?	Yes	Choose longest prefix match
Choose longest prefix match	connects to	If needed, compare administrative distance
If needed, compare administrative distance	connects to	If needed, compare metric
If needed, compare metric	connects to	Forward to next hop / exit interface

This diagram gives students a simple route-selection order before they study static routes, default routes, and floating static routes.

DEFINITION AND TECHNICAL IN-DEPTH KNOWLEDGE

17.1 Routing

Routing is the process of forwarding packets from one IP network to another. A router receives a packet, examines the destination IP address, checks its routing table, selects the best matching route, and forwards the packet toward the destination.

In Lesson 15, inter-VLAN routing introduced the idea that devices in different VLANs need Layer 3 routing to communicate. A host sends remote traffic to its default gateway. That gateway may be a router subinterface, firewall interface, or Layer 3 switch SVI. This lesson expands that idea beyond VLANs and explains how routers make forwarding decisions across many networks.

Routing exists because the world is not one flat IP network. An enterprise may have multiple VLANs, branch offices, data centers, cloud networks, VPN users, internet links, and service provider connections. Each network has its own IP prefix. Routers connect those networks and decide where traffic should go.

At Layer 2, switches forward Ethernet frames using MAC addresses. At Layer 3, routers forward IP packets using destination IP addresses and routing tables. The router removes the incoming Layer 2 frame, keeps the Layer 3 packet, makes a routing decision, and then encapsulates the packet into a new Layer 2 frame for the next hop.

For CCNA and CompTIA Network+, students must understand routing tables, connected routes, static routes, default routes, administrative distance, longest-prefix match, route selection, and floating static routes.

17.2 Technical and Business Importance of Routing

Routing matters because almost every enterprise service depends on traffic crossing networks. A staff PC may need to reach a server VLAN. A branch office may need to reach headquarters. A company firewall may need to send internet traffic to an ISP. A cloud VPN may need to route traffic between an office subnet and a cloud subnet. If routing is wrong, the physical network may be working but applications still fail.

Routing also matters for troubleshooting. When a host can reach its gateway but not a remote network, the next question is often: does the router know how to reach the destination, and does the return path exist? Many beginners test only one direction, but routing must work both ways. A packet must reach the destination, and the reply must return.

Routing is also a security and design topic. A route can make a network reachable. A missing route can isolate a network. A wrong default route can send traffic to the wrong place. A broad route can expose more networks than intended. Firewalls and ACLs control traffic, but routing determines whether a path exists.

17.3 Router Packet Forwarding Process

A router forwards packets by following a structured process.

First, the router receives a frame on an interface. The frame is a Layer 2 container, such as an Ethernet frame. The router checks whether the frame is addressed to its local interface MAC address or accepted Layer 2 destination.

Second, the router removes the Layer 2 header and trailer. This is decapsulation at Layer 2. The router then examines the Layer 3 IP packet.

Third, the router reads the destination IP address in the packet.

Fourth, the router checks the routing table for the best matching route.

Fifth, the router identifies the outgoing interface and next-hop address.

Sixth, the router builds a new Layer 2 frame for the next link. If the next link is Ethernet, the router may use ARP for IPv4 or Neighbor Discovery for IPv6 to find the next-hop MAC address.

Seventh, the router forwards the new frame out the selected interface.

Router Forwarding Process

Step	Router Action
1	Receives Layer 2 frame
2	Removes Layer 2 header/trailer
3	Reads destination IP address
4	Searches routing table
5	Selects best route
6	Identifies next hop and outgoing interface
7	Re-encapsulates packet into new Layer 2 frame
8	Forwards frame out selected interface

This is why MAC addresses change at every routed hop, while the source and destination IP addresses usually remain the same unless NAT is used.

17.4 Routing Table Components

A routing table is the database a router uses to decide where to forward packets. It contains routes to networks and instructions for reaching them.

On Cisco IOS, a routing table may be viewed with:

Command / configuration block

```
show ip route
```

A route usually contains:

- Route source or code.
- Destination network/prefix.
- Administrative distance.
- Metric.
- Next-hop address.
- Outgoing interface.

Example route:

Command / configuration block

```
S    192.168.50.0/24 [1/0] via 10.0.0.2
```

This means:

Part	Meaning
`S`	Static route
`192.168.50.0/24`	Destination network
`[1/0]`	Administrative distance 1, metric 0
`via 10.0.0.2`	Next-hop IP address

Another example:

Command / configuration block

```
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0
```

`C` means connected route. `L` means local route for the router's own interface IP address.

Common Cisco Route Codes

Code	Meaning
`C`	Connected route
`L`	Local route
`S`	Static route
`R`	RIP route
`O`	OSPF route
`D`	EIGRP route
`B`	BGP route

Dynamic routing protocols will be introduced in Lesson 17. In this lesson, the focus is connected, local, static, default, and floating static routes.

17.5 Connected and Local Routes

A connected route is automatically added to the routing table when a router interface has an IP address and is up/up. It represents the network directly attached to that interface.

Example:

Command / configuration block

```
interface gigabitEthernet 0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
```

The router adds:

Command / configuration block

```
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
```

The connected route tells the router that the `192.168.10.0/24` network is directly reachable through `G0/0`. The local route identifies the router's own interface address.

Connected routes matter because they are the foundation of routing. A router knows directly connected networks automatically, but it does not automatically know remote networks unless a route is configured or learned.

17.6 Static Routes

A static route is a route manually configured by an administrator. Static routes tell the router how to reach a network that is not directly connected.

Static routes exist because some networks are simple enough that manual routing is practical. A small branch router may need only a default route to the ISP. A lab router may need a few static routes. A firewall may use static routes to reach internal networks. Static routes are predictable and do not require a dynamic routing protocol.

Basic Cisco IOS static route syntax:

Command / configuration block

```
ip route <destination-network> <subnet-mask> <next-hop-ip>
```

Example:

Command / configuration block

```
ip route 192.168.50.0 255.255.255.0 10.0.0.2
```

This tells the router:

Command / configuration block

```
To reach 192.168.50.0/24, send traffic to next-hop 10.0.0.2.
```

Static route using outgoing interface:

Command / configuration block

```
ip route 192.168.50.0 255.255.255.0 gigabitEthernet 0/1
```

Static route using both next hop and outgoing interface:

Command / configuration block

```
ip route 192.168.50.0 255.255.255.0 gigabitEthernet 0/1 10.0.0.2
```

In Ethernet networks, using a next-hop IP address is usually preferred because the router can resolve the next hop properly. Point-to-point links may use outgoing interface syntax more safely.

Static Route Components

Component	Example	Purpose
Destination network	`192.168.50.0`	Remote network to reach
Subnet mask	`255.255.255.0`	Defines destination prefix
Next hop	`10.0.0.2`	Router to forward traffic to
Outgoing interface	`G0/1`	Interface used to send traffic

17.7 Static Route Example

Consider two routers.

Command / configuration block

```
LAN A --- R1 --- 10.0.0.0/30 --- R2 --- LAN B
```

Addressing:

Network	Details
LAN A	192.168.10.0/24
R1 LAN IP	192.168.10.1
R1 WAN IP	10.0.0.1/30
R2 WAN IP	10.0.0.2/30
R2 LAN IP	192.168.20.1
LAN B	192.168.20.0/24

R1 knows `192.168.10.0/24` and `10.0.0.0/30` because they are connected. R1 does not automatically know `192.168.20.0/24`.

Configure R1:

Command / configuration block

```
ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

R2 knows `192.168.20.0/24` and `10.0.0.0/30` because they are connected. R2 does not automatically know `192.168.10.0/24`.

Configure R2:

Command / configuration block

```
ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

This creates two-way reachability.

Important troubleshooting rule:

Command / configuration block

```
Routing must work in both directions.
```

If R1 has a route to LAN B but R2 has no route back to LAN A, the ping may fail because the reply cannot return.

17.8 Default Routes

A default route is a route used when no more specific route matches the destination. It is often called the gateway of last resort.

Default route syntax in IPv4:

Command / configuration block

```
ip route 0.0.0.0 0.0.0.0 <next-hop-ip>
```

Example:

Command / configuration block

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

This tells the router:

Command / configuration block

```
If you do not have a more specific route, send the packet to 203.0.113.1.
```

Default routes are common at the internet edge. A branch router may not need routes for every internet destination. It can send unknown destinations to the ISP using a default route.

Example:

Command / configuration block

```
Branch LAN -> Branch Router -> ISP Router -> Internet
```

The branch router may have:

Command / configuration block

```
ip route 0.0.0.0 0.0.0.0 <ISP-next-hop>
```

Default Route Summary

Feature	Description
IPv4 default network	`0.0.0.0`
IPv4 default mask	`0.0.0.0`
CIDR notation	`0.0.0.0/0`
Purpose	Match all destinations not matched by a better route
Common use	Internet edge, branch routers, stub networks

For IPv6, the default route is:

Command / configuration block

```
ipv6 route ::/0 <next-hop-ipv6>
```

17.9 Routing Table Lookup and Longest-Prefix Match

When a router receives a packet, it chooses the most specific matching route. This is called longest-prefix match.

Longest-prefix match means that if multiple routes match the destination IP address, the router chooses the route with the longest prefix length, because it is the most specific.

Example routing table:

Route	Meaning
`0.0.0.0/0`	Default route
`10.0.0.0/8`	Broad route for 10.x.x.x
`10.10.0.0/16`	More specific route
`10.10.20.0/24`	Most specific route

If the destination is:

Command / configuration block

```
10.10.20.55
```

All four routes may match, but the router chooses:

Command / configuration block

```
10.10.20.0/24
```

because `/24` is the longest prefix.

Longest-Prefix Match Example

Destination IP	Matching Routes	Chosen Route
10.10.20.55	`/0`, `/8`, `/16`, `/24`	`10.10.20.0/24`
10.10.99.5	`/0`, `/8`, `/16`	`10.10.0.0/16`
10.50.1.10	`/0`, `/8`	`10.0.0.0/8`
8.8.8.8	`/0`	`0.0.0.0/0`

This rule is extremely important. A default route is used only when no more specific route exists.

17.10 Administrative Distance

Administrative distance, or AD, is a trust value used by Cisco routers to choose between routes to the same destination learned from different sources.

Lower administrative distance is preferred.

Example:

Command / configuration block

```
Static route to 192.168.50.0/24
OSPF route to 192.168.50.0/24
```

If both routes describe the exact same prefix, the router compares administrative distance. A static route normally has AD 1. OSPF normally has AD 110. The static route is preferred because 1 is lower than 110.

Common Administrative Distance Values

Route Source	Administrative Distance
Connected	0
Static	1
EIGRP summary	5
External BGP	20
Internal EIGRP	90
OSPF	110
RIP	120
External EIGRP	170
Internal BGP	200
Unknown/unusable	255

Administrative distance is considered only after longest-prefix match determines comparable routes. A more specific route usually wins over a less specific route even if the less specific route has a better AD.

17.11 Route Selection Order

Route selection can be understood in a simple order.

First, the router looks for matching routes in the routing table.

Second, it chooses the longest prefix match. The most specific route wins.

Third, if multiple routes have the same prefix length and destination, the router chooses the route with the lowest administrative distance.

Fourth, if administrative distance is tied, the router uses the metric to choose the best route.

Fifth, if equal-cost routes remain, the router may load balance depending on platform and configuration.

Route Selection Order

Order	Decision Factor	Preferred Value
1	Longest prefix match	Most specific prefix
2	Administrative distance	Lowest AD
3	Metric	Lowest metric
4	Equal-cost paths	May load balance

Students should avoid a common misunderstanding: administrative distance does not override longest-prefix match. A `/24` route is more specific than a `/16` route. The router uses the most specific matching route.

17.12 Floating Static Routes

A floating static route is a static route configured with a higher administrative distance so it acts as a backup route.

Normal static routes have AD 1. If we configure a static route with AD 200, it will not be used while a better route exists. It "floats" in the background and becomes active only if the primary route disappears.

Example:

Primary route through ISP1:

Command / configuration block

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

Backup floating static route through ISP2:

Command / configuration block

```
ip route 0.0.0.0 0.0.0.0 198.51.100.1 200
```

The first default route has AD 1. The second default route has AD 200. The router uses ISP1 while it is available. If the primary route is removed from the routing table, the backup route can be installed.

Floating Static Route Summary

Route	Next Hop	AD	Role
`0.0.0.0/0`	203.0.113.1	1	Primary
`0.0.0.0/0`	198.51.100.1	200	Backup

Floating static routes are useful for backup internet links, backup WAN links, and simple failover designs. However, they only work if the primary route is actually removed from the routing table. In some cases, tracking or dynamic routing may be needed for better failure detection.

17.13 Static Routing Design Use Cases

Static routing is useful in specific situations.

Common use cases include:

Use Case	Why Static Routing Fits
Stub network	Only one way out
Internet edge default route	Send unknown destinations to ISP
Small branch	Few networks, simple design
Lab environment	Predictable and easy to learn
Firewall routing	Explicit paths to internal networks
Backup path	Floating static route

Static routing is not always the best solution. In large networks with many routes and changing paths, dynamic routing is easier to scale. Lesson 17 will introduce dynamic routing concepts.

17.14 Common Static Routing Mistakes

Static route mistakes are common because manual routes require exact information.

One mistake is using the wrong next-hop IP address. The next hop must be reachable from the router.

Another mistake is forgetting the return route. If R1 can reach R2's LAN but R2 does not know how to return traffic to R1's LAN, communication fails.

Another mistake is using an incorrect subnet mask. A route to `192.168.10.0 255.255.255.0` is different from a route to `192.168.10.0 255.255.0.0`.

Another mistake is relying on a default route when a more specific route is required through another path.

Common Static Route Mistakes

Mistake	Result
Wrong next hop	Route may fail or traffic goes wrong direction
No return route	Replies cannot return
Wrong mask	Too much or too little traffic matches
Missing default route	Unknown destinations fail
Floating route AD too low	Backup route becomes primary
Interface down	Connected and static routes may disappear

17.15 Routing Troubleshooting

Routing troubleshooting should be structured and directional.

A practical sequence is:

1. Confirm the source host IP address, mask, and gateway.
2. Confirm the destination IP address.
3. Test the source host's default gateway.
4. Check the first router's routing table.
5. Confirm the router has a route to the destination network.
6. Confirm the next hop is reachable.

7. Check the return path from the destination side.

8. Check ACLs, firewall policy, or NAT if routing appears correct.

9. Use traceroute to identify where the path stops.

10. Check interface status and counters.

Routing Troubleshooting Table

Symptom	Possible Cause	What to Check
Host cannot ping gateway	Local VLAN/subnet/gateway issue	Host IP, gateway, switch/SVI
Router has no route	Missing static/default route	`show ip route`
Route exists but next hop unreachable	Link or addressing problem	`ping <next-hop>`, interface status
One-way communication	Missing return route	Remote router routing table
Internet unreachable	Missing/wrong default route	`0.0.0.0/0`, ISP next hop
Backup link used unexpectedly	AD or primary route issue	Static route AD, interface status
Traceroute stops at router	Missing next route or policy block	Routing table, ACL/firewall

17.16 Windows, Linux, and Cisco IOS Practical Commands

Windows Commands

Command / configuration block

```
ipconfig /all
route print
ping <default-gateway>
ping <remote-destination>
tracert <remote-destination>
```

Windows helps verify the host's local route, default gateway, and path to a remote destination. `route print` shows the local host routing table.

Linux Commands

Command / configuration block

```
ip addr
ip route
ping <default-gateway>
ping <remote-destination>
traceroute <remote-destination>
```

Linux shows routes clearly with `ip route`. A missing `default via` route means the host may not know how to reach remote networks.

Cisco IOS Commands

Command / configuration block

```
show ip interface brief
show ip route
show ip route static
show ip route connected
show running-config | include ^ip route
ping <destination>
traceroute <destination>
show arp
```

These commands verify interface status, route sources, static route configuration, next-hop reachability, and forwarding path.

17.18 Cybersecurity Relevance

Routing affects security because it determines reachability. If a route exists, traffic may be able to reach a network unless policy blocks it. If a sensitive network is accidentally advertised or statically routed where it should not be, exposure increases.

Static routes should be documented and reviewed. A route to a management subnet, server subnet, or partner network should not be added casually. Routes should match business need and security policy.

Attackers may also benefit from misrouting. If traffic is sent to the wrong next hop, it may be intercepted, blackholed, or bypass expected inspection points. In secure designs, routing should guide traffic through firewalls, IDS/IPS, logging points, or zero-trust enforcement where required.

Security principle:

Command / configuration block

```
Routing creates paths. Security policy controls who may use those paths.
```

17.19 Cloud Networking Relevance

Cloud networking uses routing tables heavily. A cloud subnet may have a route table that sends local traffic inside the virtual network, internet traffic to an internet gateway, private outbound traffic to a NAT gateway, and corporate traffic to a VPN or cloud firewall.

Example cloud route table:

Destination	Target
10.100.0.0/16	Local cloud network
0.0.0.0/0	Internet gateway
10.10.0.0/16	VPN gateway
10.20.0.0/16	Cloud firewall

The same concepts apply: longest-prefix match, default route, next hop, and policy. Cloud engineers must also avoid overlapping routes between office networks, cloud VPCs/VNets, VPNs, and partner networks.

18. KEY TERMS / GLOSSARY

Term	Meaning
Routing	Forwarding packets between IP networks.
Routing Table	Database used by a router to select forwarding paths.
Route	Entry that tells a router how to reach a destination network.
Next Hop	The next router IP address where traffic should be sent.
Outgoing Interface	Interface used to forward traffic.
Connected Route	Automatically learned route for a directly connected network.
Local Route	Route for the router's own interface IP address.
Static Route	Manually configured route.
Default Route	Route used when no more specific route matches.
Gateway of Last Resort	Cisco term for the default route.
Longest-Prefix Match	Route selection rule choosing the most specific matching prefix.
Administrative Distance	Trust value used to choose between routes from different sources.
Metric	Value used by a routing source to choose best path among similar routes.
Floating Static Route	Static route with higher AD used as a backup route.
Return Path	Route used by reply traffic to get back to the source.

19. COMPARISON TABLES

19.1 Connected vs Static vs Default Routes

Route Type	How Created	Common Purpose
Connected	Interface up with IP address	Reach directly attached network
Static	Manually configured	Reach specific remote network
Default	Manually or dynamically learned	Send unknown destinations to next hop

19.2 Longest-Prefix Match vs Administrative Distance

Feature	Longest-Prefix Match	Administrative Distance
Purpose	Chooses most specific matching route	Chooses trusted source for same prefix
Preferred Value	Longer prefix	Lower AD
Example	`/24` beats `/16`	Static AD 1 beats OSPF AD 110
Common Confusion	Happens before AD comparison	Does not override specificity

19.3 Static Route vs Floating Static Route

Feature	Static Route	Floating Static Route
Administrative Distance	Usually 1	Manually set higher
Role	Primary manual path	Backup manual path
Installed When	Next hop/interface valid	Better route unavailable
Use Case	Small network route	Backup ISP/WAN link

19.4 Routing Troubleshooting Tools

Tool/Command	Purpose
`ping`	Tests reachability
`traceroute` / `tracert`	Shows path and where traffic stops
`show ip route`	Displays router routing table
`show ip interface brief`	Shows interface IP and status
`show arp`	Shows IPv4-to-MAC mappings
`route print`	Shows Windows routing table
`ip route`	Shows Linux routing table

20. REAL-WORLD EXAMPLE / ANALOGY

Routing is like a delivery company choosing roads to reach different towns. The routing table is the driver's map. A specific route to one town is better than a broad instruction such as "send everything north." If the driver has no specific route, the default route is the general road used for unknown destinations.

Administrative distance is like trust in the source of the map. A directly observed road is trusted most. A manually approved route is highly trusted. A route learned from another system may be trusted based on the protocol.

21. QUICK-RECAP CHEAT SHEET

- Routers forward packets between IP networks.
- Routers use destination IP addresses.
- Routers consult the routing table.
- Connected routes appear when interfaces are up with IP addresses.
- Local routes represent the router's own interface IP addresses.
- Static routes are manually configured.
- Default routes match unknown destinations.
- IPv4 default route is `0.0.0.0/0`.
- Cisco default route syntax is `ip route 0.0.0.0 0.0.0.0 <next-hop>`.
- Longest-prefix match chooses the most specific route.
- Administrative distance chooses between routes to the same prefix from different sources.
- Lower administrative distance is preferred.
- Connected AD is 0.
- Static AD is 1.
- OSPF AD is 110.
- RIP AD is 120.
- Floating static routes use higher AD as backup routes.
- Routing must work in both directions.
- `show ip route` is the main routing table command.
- `traceroute` helps locate where the path stops.

7. LESSON ASSESSMENT

This assessment is intentionally focused. Students should answer clearly, use correct networking terminology, and show practical troubleshooting thinking.

7.1 Theory Questions

1. Explain the main networking concept covered in this lesson and describe its role in a real enterprise network.

2. Describe two technical problems that can occur when this lesson's concept is poorly designed or incorrectly configured.

3. Explain how a network technician or junior network engineer would verify that this part of the network is working correctly.

4. Describe one security, reliability, or performance concern related to this topic.

5. Explain how this topic connects to CCNA, Network+, IT support, or junior network engineering work.

7.2 Multiple-Choice Questions

1. What is the best reason for studying this lesson's topic?

A. It helps students understand and support real network operation. B. It replaces the need for troubleshooting. C. It removes the need for IP addressing. D. It is only useful for memorizing definitions.

2. When troubleshooting a problem related to this lesson, what should be done first?

A. Gather evidence and identify the exact scope of the problem. B. Replace every network device immediately. C. Delete the running configuration without backup. D. Assume the Internet service provider is always responsible.

3. Which habit best reflects professional network engineering practice?

A. Verify changes with commands, tests, and documented results. B. Make random changes until something works. C. Ignore diagrams and rely only on memory. D. Avoid documenting successful fixes.

4. Which item is most useful when proving that a network feature is working?

A. A successful test result compared with the expected result. B. A guess from another user. C. A device name with no configuration evidence. D. A screenshot unrelated to the problem.

5. What should students connect this lesson to during hands-on practice?

A. Addressing, device roles, commands, verification, and troubleshooting. B. Only typing speed. C. Only cable color. D. Only memorizing vendor logos.

6. In an enterprise environment, why is documentation important for this topic?

A. It helps future troubleshooting, audits, handover, and change control. B. It makes testing unnecessary. C. It disables security risks automatically. D. It replaces the need for correct configuration.

7. Which approach gives the strongest learning outcome?

A. Combining technical explanation with lab practice and verification. B. Reading commands without understanding them. C. Skipping troubleshooting steps. D. Memorizing answers without using the network.

8. If a configuration change causes a new outage, what should the technician use?

A. A rollback plan and saved baseline information. B. A random new subnet. C. A different monitor. D. A guess without evidence.

9. Which result shows that a student understands the lesson at a practical level?

A. The student can explain the concept, configure or identify it, test it, and troubleshoot failure. B. The student can only repeat the title. C. The student avoids using commands. D. The student ignores enterprise examples.

10. Why are scenario questions included in this course?

A. They train students to apply concepts to real network problems. B. They remove the need for labs. C. They focus only on spelling. D. They are unrelated to IT jobs.

7.3 Scenario-Based Questions

1. A user or department reports a connectivity problem related to this lesson's topic. Write the first five checks you would perform, the commands or tools you would use, and the evidence you expect to collect. 2. You are asked to explain this topic to a new IT support intern before a hands-on lab. Write a short technical explanation that connects the concept to real devices, real traffic flow, and troubleshooting.

7.4 Answer Guide

Theory answers should be marked for technical accuracy, clear explanation, correct terminology, and practical relevance to enterprise networking. Scenario answers should show a logical troubleshooting sequence, evidence gathering, likely causes, verification steps, and a professional final recommendation.

Multiple-choice answers: 1. A; 2. A; 3. A; 4. A; 5. A; 6. A; 7. A; 8. A; 9. A; 10. A